

Computer Architecture And Organization Mid T2.(2022-2023)

Q1) (5 Points) Express The Following HexaDecimal To Decimal Using IEEE 754 Floating Point Representation .

0x40F00000	0XC1400000

Q2) (5 Points) Based On The Final Division Algorithm , Complete 0101(two)/0011(two) By Filling In The Following Table .

<u>Iteration</u>	<u>Divisor (D)</u>	<u>Reminder (R)</u>	<u>Remark</u>

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Q3) (10 Points) For Each Instruction In The Following Code Segment, Determine The Instruction Format, And The Decimal And Values Of Each Instruction Field .

and \$s1 , \$zero , \$zero

add \$t0 , \$s0 , \$zero

loop : lw \$t1 , 0(\$t0)

sub \$s1 , \$s2 , \$t1

addi \$t0 , \$t9 , -4

bne \$t1 , \$zero , loop

Memory Address	Machine Code	Instruction Format

Instruction	Code
Add	32
Addi	8
And	36
Lw	35
Bne	5
Sub	34

Register Name	Register Number
\$t0-\$t7	8-15
\$s0-\$s7	16-23
\$zero	0

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Q4) (10 Points) In MIPS Assembly, Write An Assembly Language Version Of The Following C Code Segment : (Add The Numbers From 0 To 9)

Int i;

Int sum=0;

for (i = 0 ; i != 10 ; i = i + 1)

sum = sum + i ;

At The Beginning Of This Code Segment, The Only Values In Registers Are :

\$*s0* = i

\$*s1* = sum